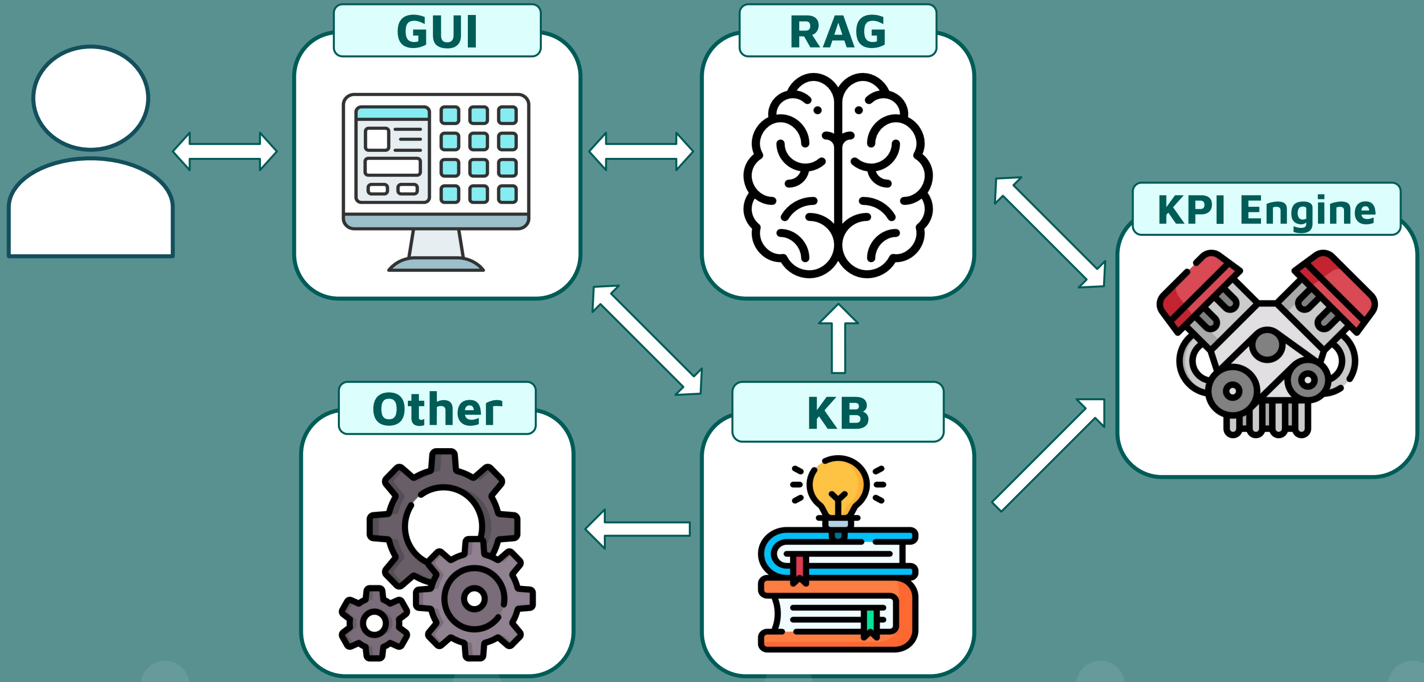
My Upgrade: Enhancing Code Structure and Integrating Processes into the Ontology

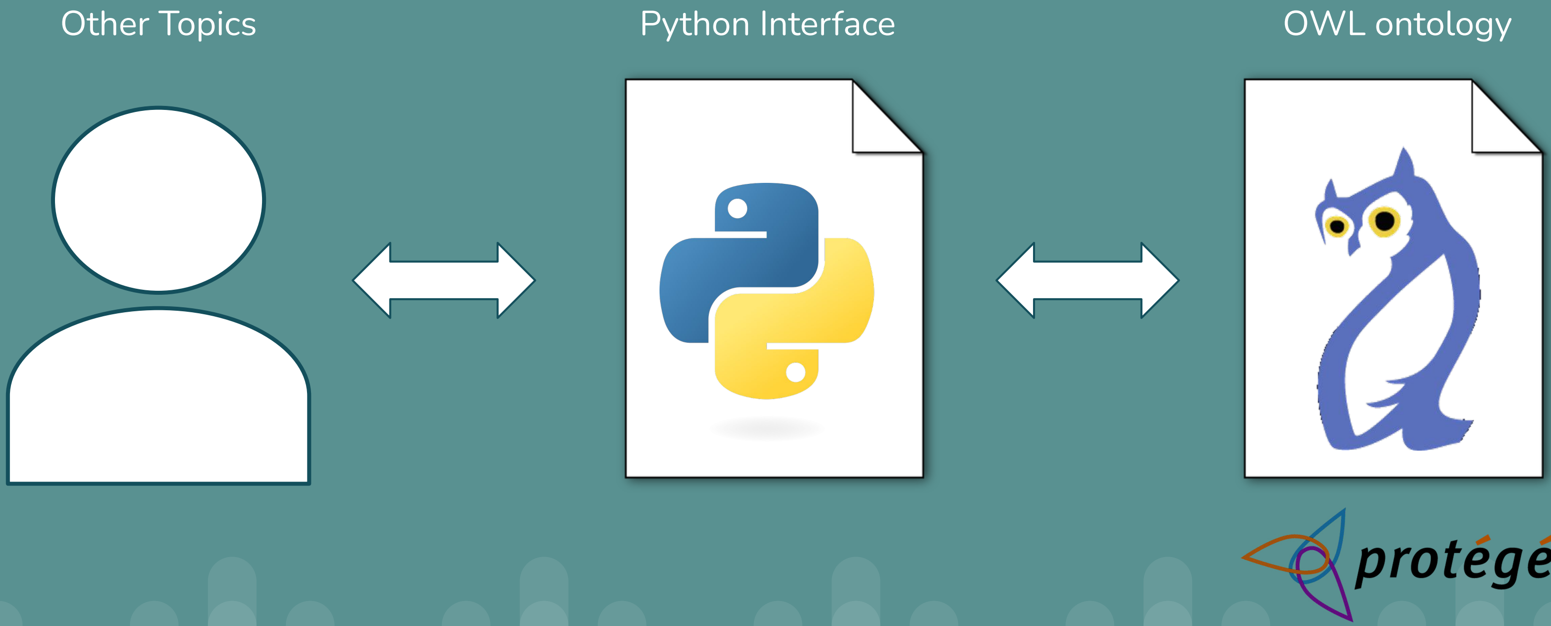
Master Degree in Computer Science, University of Pisa
Smart Applications
Academic Year 2024/2025

Giacomo Aru - g.aru@studenti.unipi.it - 597700

Topic 1: Role inside the Project



Topic 1: Final Deliverables

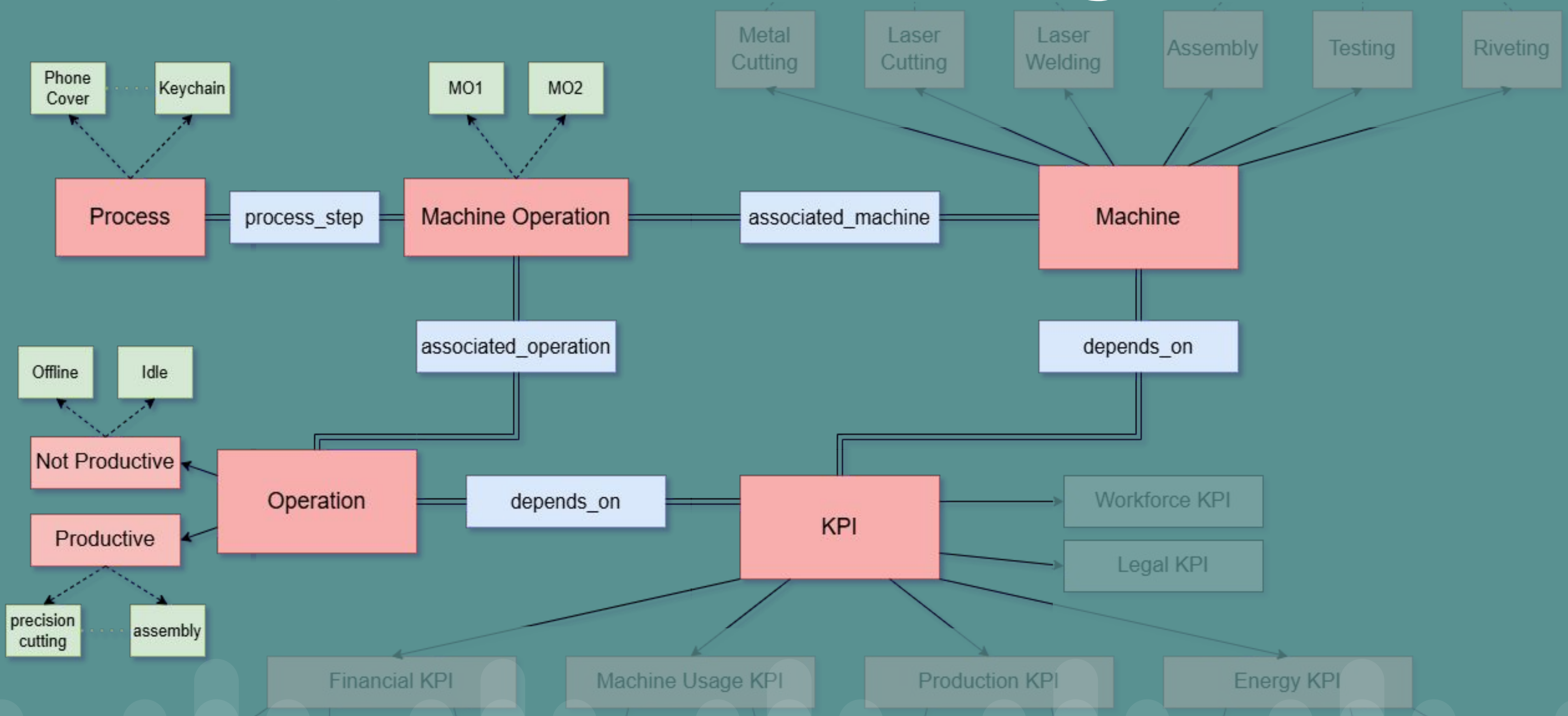


My Upgrade: Process Integration

A **process** is defined as a **series of machine-operation pairs** encompassing all the steps required to **produce** a **finished product**

```
process_1 = {
    "process_label": "phone_cover_creation",
    "process_description": "Production process for a phone cover,
    including cutting, welding, assembly, and quality testing.",
    "steps_list": [
        ("medium_capacity_cutting_machine_1", "material_cutting"),
        ("laser_welding_machine_1", "edge_welding"),
        ("assembly_machine_2", "assembly"),
        ("testing_machine_1", "quality_testing")
    ]
}
```

My Upgrade: Process Integration



My Upgrade: Process Integration

```
def add_process(process_label, process_description, steps_list):
    # Adds a process to the KB

def delete_process(process_label):
    # Removes a process from the KB

def get_steps(process_label):
    # Retrieves steps for the specified process

def get_closest_process_steps(process_label, method='levenshtein'):
    # Retrieves steps of the closest matching process
```

My Upgrade: Enhanced Dynamicity

```python
def add_kpi(superclass, label, description, unit_of_measure,
    parsable_computation_formula, human_readable_formula=None,
    depends_on_machine=False, depends_on_operation=False):
    # Adds a KPI to the system

def delete_kpi(label):
    # Deletes a KPI from the system

def add_operation(label, description):
    # Adds an operation to the system

def delete_operation(label):
    # Deletes an operation from the system
```

My Upgrade: Refined Function and Error Handling

```
<<< kbi.get_closest_class_instances('class_label', istances_type='a')
# max similarity label -> laser_welding_machine
>>>(['laser_welding_machine_1', 'laser_welding_machine_2'], 0.488)

<<< kbi.get_closest_class_instances('class_label', istances_type='k')
# max similarity label -> machine_usage_kpi
>>>(['mean_time_between_failures', ... , 'time_max'], 0.485)

<<< kbi.get_closest_class_instances('class_label', istances_type='p')
# max similarity label -> process
>>>(['customized_keychain_production', ... ], 0.25)

<<< kbi.delete_operation('riveting_assembly')
>>> ValueError: The operation 'riveting_assembly' cannot be deleted
    because it is used in the process step: ...
```

My Upgrade: Custom Similarity Measure

```
<<< kbi.get_closest_object_properties('cycles', method='levenshtein')
>>> ({'label': 'cycles_kpi', ... }, 0.6)

<<< kbi.get_closest_object_properties('cycles', method='custom')
>>> ({'label': 'cycles_sum', ...}, 0.8500000015)
```

0.85 0.15

kpi_label_agg -> kpi_label + agg -> kpilabel + agg

Thanks for your attention

Master Degree in Computer Science, University of Pisa
Smart Applications
Academic Year 2024/2025

Giacomo Aru - g.aru@studenti.unipi.it - 597700